Level Data

Level Data Accessibility

Testing Process and Prerequisites

Version 1.1 • Prepared by Carly Falcioni and Galen Scott

Contents

- Overview/Purpose of Slide Deck
- Tools
  - Purpose/Use Description
  - Testing Guides
- Accessibility testing - What is involved
  - Expectations & Preparation
  - Evaluating needs
  - Estimating scope of work
  - Planning the work
- Accessibility - Doing the work (in a sprint)
  - High level play-by-play

Level Data

Tools for Accessibility Testing

Accessibility Tools

LevelData

There are a wide variety of tools that are used for testing Accessibility.

Testing Tool / Method	Purpose / What It Does
✅ **Automated Scanning Tool**	Scans pages for predicted accessibility issues. Note: does **not** replace manual testing.
🗣 **Screen Reader**	Reads out content from or about the page to assist visually impaired users.
⌨️ **Keyboard Navigation**	Ensures all content is accessible and usable without a mouse.
📱 **Touchscreen Interaction**	Validates content is accessible for users on touch devices.
🔧 **Element Inspector Tool**	Allows you to inspect individual page elements for accessible markup and structure.
🎨 **Contrast Checker**	Tests the contrast ratio between text and background to ensure visibility for low-vision users.

Testing Guides & Key Points (1 of 2)

LevelData

Automated Scans

Objectives of testing

Ensure:

- Non-visual content conforms to accessibility standards.
- You re-scan the page every time a change occurs on the page including when:
  - modal opened
  - dropdown item selected
  - links or buttons become enabled/disabled

TIP: Keep the dev tools bar docked to the bottom of your screen, not the side (highly recommended).

- This ensures the page is altered as little as possible and stays in desktop view.

Screen Reader

Objectives of testing

Ensure:

- Page content is announced in a logical order.
- Images essential to operation of the page have alternate text that the screen reader announces.
- Identifiable information such as full names, units of measurement, and categories are announced.

Examples:

"Column 5 of 9" is not sufficient.
✅ "Column 5 of 9 Academic Arts High School" is sufficient.

"Remove widget" is not sufficient.
✅ "Remove notification widget" is sufficient.

"10.5" or "3" is not sufficient.
✅ "10.5 percent" and "3 percent" are sufficient.

Keyboard Only

Objectives of testing

Ensure:

- All functionality that can be completed with a mouse, is also operable by using the keyboard only.

Ensure:

- The focus indicator...
  - moves in a logical order
  - only tabs to interactive elements
  - does not get trapped on any elements.

TIP: If you noticed the tab order was not logical or visible, you can use the ANDI bookmarklet (next slide) to visually view the exact tab order by clicking the "tab order" link.

Testing Guides & Key Points (2 of 2)

LevelData

Touch Screen	ANDI Bookmarklet	WebAIM Contrast Checker
Objectives of testing Ensure: ○ The flow of work is not disrupted when alternating from one input method to another. ○ Ex: Use a touch screen to open a dropdown, then use the keyboard to make your selection. The focus indicator and flow of work should not be affected by alternating input methods.	Objectives of testing Ensure: ○ The elements and structure of the page have accessible markup by selecting the type of content you want to evaluate and hovering over the element you want to inspect. TIP: If you need to inspect an issue at 200% zoom, turn on the "Minimode" display in ANDI so the menu takes up less space on the page.	Objectives of testing Ensure: • The contrast of elements on the page are meeting accessibility standards by entering the HEX code of the background and foreground into the tool or using the eyedropper to hover and select the elements you want to compare.

The Importance of Manual Testing

LevelData

Automated testing CAN...

- Make predictions about ARIA attributes, keyboard tab stops, and structural issues.

Automated testing CAN'T:

- Interpret context
  - Determine whether content is decorative or essential
  - Assess whether text is meaningful and understandable
  - Assess whether abbreviations/unique words are pronounced correctly
- Test concurrent input mechanisms
  - Switching between methods of input, such as touchscreen and keyboard use during a workflow
- Identify and associate equivalent controls
  - Determine if there is more than one way to accomplish the same task on a given page

Level Data

Accessibility testing for

a product:

Preparation & Expectations

Product Testing

Evaluate the needs

Level Data

Before work can begin...

- The following form detailing the expected conformance level and other information must be completed and provided to the accessibility testers
  - Accessibility Form - Bringing a Product Up to Standard

PURPOSE: This ensures time is spent only testing applicable/necessary WCAG Success Criteria.

- Testers must be informed whether the testing is intended for a Voluntary Product Accessibility Template (VPAT) or as part of general maintenance/regression testing.

PURPOSE: This information determines timeline formation and testing approach.

Estimate scope of work (1 of 2)

LevelData

Preparation + Readiness

- Testers are given access to the product.
- Testers are given an up-to-date, recorded walkthrough of the product (either live or previously documented).
- Testers are given at least one week to familiarize with the product and the targeted workflows to be tested.

Why?

- This gives testers the insight needed to form a timeline that is both accurate and realistic.
- The more familiar testers are with the product, the more efficiently they can plan their approach for testing.

Estimate scope of work (2 of 2)

LevelData

What insight can help determine a timeline?

- Was the product built with any capacity of accessibility?
- How many pages are there to the product?
- How many varying states of pages are there? (enabled/disabled elements, returned states, etc)
- When are modals triggered in the product? (warnings, alerts, input fields, etc)
  - Some modals only show up under certain circumstances and thus can be easily overlooked by testers who've had limited exposure to the product. This information ensures every component is tested and accounted for.
- What kind of components does the product contain?
  - Ex: Simple content such as standard text and images only (articles/informative content.)
  - Ex: Complex content such as input forms, charts, calendar pop-outs, grids, timed content, etc.
- Does the product contain any third-party components? (These are exempt from testing)
- What kind of documentation can be generated from the site? (PDFs, CSVs, etc)
- How many places in the product can documentation be generated?

Plan the work

- Discuss with the team which approaches to testing might work most efficiently for the situation at hand:
    A. Assign each tester one type of testing, and everyone tests the same page.
    B. Assign each tester a different page, of which they each employ all types of testing.
    C. Whatever works! Brainstorm as a team on what makes the plan feel feasible and achievable.

Pros and Cons:

- Option A achieves broad coverage of WCAG Success Criteria faster but slows down coverage of the product
- Option B achieves broad coverage of the product faster but dramatically slows testing results.

- Overall: Option A will be more thorough, organized, and efficient.
- Overall: Option B will be less organized, less time efficient, but will saturate the product faster.

Level Data

Accessibility testing for

a product:

What is involved?

Product Testing

Complex Testing

LevelData

- Testing all pages in both 200% view and 400% view
- Testing with at least one screen reader
  - Using more than one is encouraged since not all screen readers will read content the same way
- Testing with a keyboard only
- Conducting these tests for every state of the pages (enabled/disabled content, modals open, returned states etc.)
- Conducting these tests for any documentation generated by the site (PDFs, CSVs, etc.)
- Conducting these tests in multiple browser types

Team Efforts

LevelData

- Both the testers and developers will need to spend sufficient time reviewing W3.org resources (see Appendix C) to determine the best approach for fixes.
  - Some resolutions may be context-specific, and some WCAG criteria list various exceptions.
- In some cases, new elements may need to be built or restructured in order to meet accessibility requirements.
- Time will be spent training team members/lending support with testing.

Impact

- The more familiar everyone is with the same source material, the easier it will be to document and communicate findings accurately, which ensures backtracking is minimized, while time spent addressing more issues is maximized.

LevelData

Tracking Findings

- Spreadsheets
- Guiding documents
  - Baseline Accessibility Testing Guide for Product Owners
  - If you are not a product owner, you can still use the above document as a framework for approaching your tests for each page

Tickets

- Should contain links to the relevant WCAG Success Criteria being impacted (specifically, the "Understanding" docs for those criteria. See Appendix C: Helpful Accessibility Resources)
- Should contain links to other relevant resources pertaining to fixing the issue
  - W3's Pattern guides, Practices guides, etc...(See Appendix C: Helpful Accessibility Resources)

Sprints

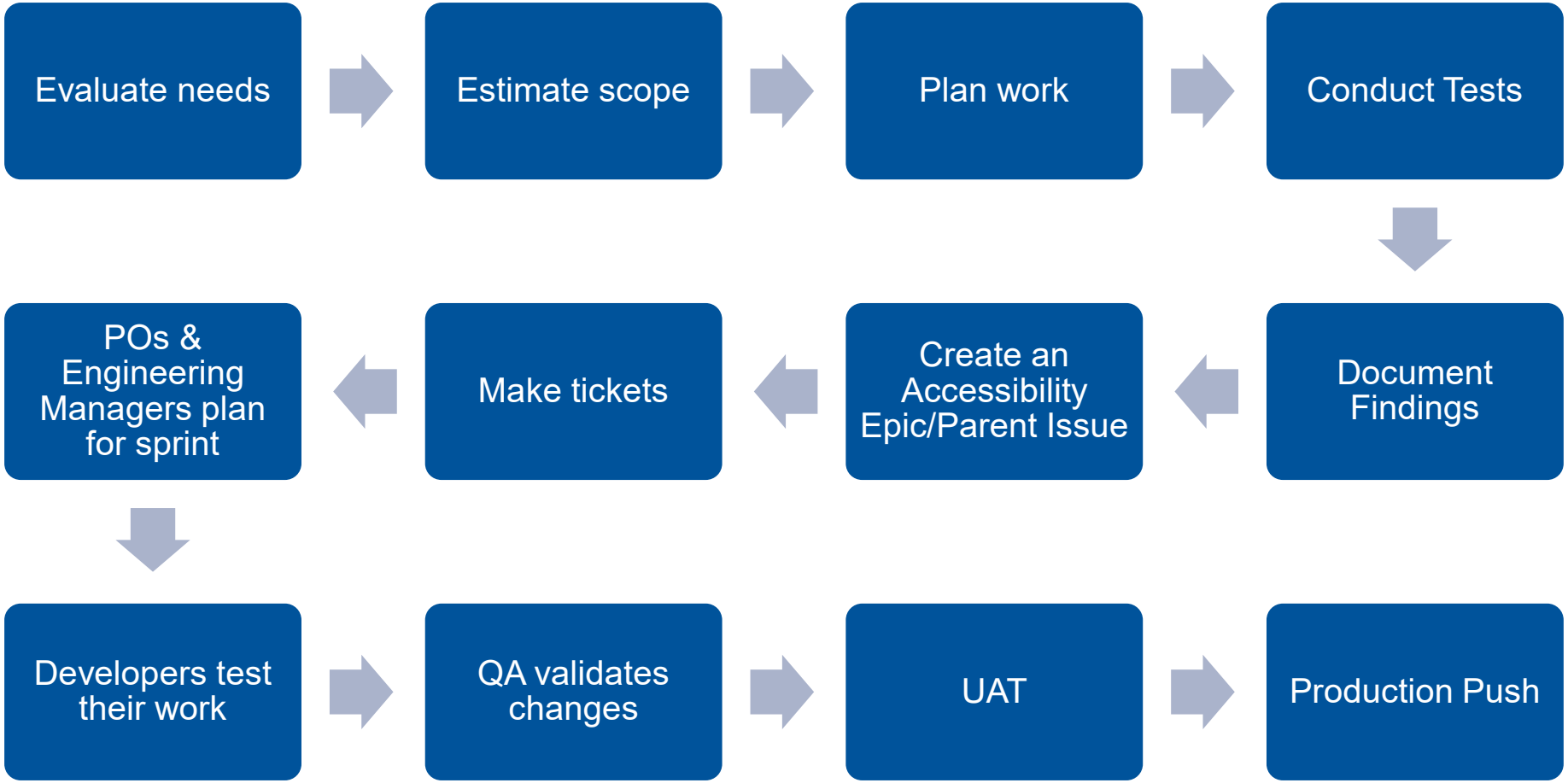
- POs create an Epic/Parent Issue for tickets to go in
  - Every Epic description contains the resources listed in Appendix C: Helpful Accessibility Resources
- POs and Engineering Managers discuss how to move forward considering:
  - Team capacity
  - Difficulty and size of work
  - Priorities of this work & other suite work

As tickets are completed...

LevelData

- Engineers use tools to confirm that their changes are working.
- QA testers take tickets to review after engineers complete work.
- POs create testing information for Customers if required (if customer involvement).
- UAT (timed to work with the other sprint work/customer work.)
- Production push

High Level Play-By-Play



Level Data

Accessibility testing for a

Voluntary Product Accessibility Template (VPAT):

Preparation & Expectations

VPAT Testing

So, what is a VPAT?

Voluntary Product Accessibility Template (VPAT) is a standardized form that vendors use to document how an information and communications technology (ICT) product or service conforms with accessibility requirements, most notably Section 508 of the Rehabilitation Act (29 U.S.C. § 794(d)). It enables federal contracting officials and other buyers to make preliminary assessments of a product's accessibility features and identify areas needing improvement.

The purpose, key objectives, and benefits of having a VPAT for a product are also detailed in our What is a VPAT? document.

Evaluate the Needs

Before any work can be done for a VPAT, we need to evaluate what the need are for the product and where the product stands with A11y (Accessibility).

- The following form has been created to consolidate the information needed to conduct a VPAT: Accessibility Form - If you need a VPAT

Estimate the scope of work

LevelData

- Same roadmap of preparations as for general testing:
  - Be informed of expected conformance level
  - Be given training and reference material
  - Familiarize with the product
  - Have documented insight about the product to form a timeline and approach

Plan the work

LevelData

- Same approach as general testing:
  - Team discusses what method may work best for their situation.
  - Confidence and comfortability are priority!

Level Data

Accessibility testing for a

Voluntary Product Accessibility Template (VPAT):

What is involved?

VPAT Testing

Conducting a VPAT involves...

LevelData

- Understanding the Expected Testing Standard
  - There is more than one VPAT type, and multiple WCAG conformance levels
- Referring to WCAG Success Criterion
  - The evaluator will review the list of Success Criterion applicable to the product and ensure those criteria are met on every page, and every state of those pages of the product.
- Testing Full Pages and Processes
  - When conducting a VPAT evaluation based upon WCAG criteria, it is expected that testing accounts for full pages, complete processes, and only accessibility-supported ways of using the product. View WCAG 2.0 Conformance Requirements for more information.
- Documenting Findings
  - It is crucial to thoroughly read the VPAT instructions and ensure a full understanding of how the response terms are interpreted.
- Using the most current VPAT template: https://www.itic.org/policy/accessibility/vpat

Keep in mind...

LevelData

Perspective is imperative

- The VPAT is not a checklist
  - Once a success criterion is met once, that does not mean it can be checked off the list and you move on to the next.

As the evaluator(s), you are testing each criterion against *every page, every component, in every state.*

- In order for a Success Criterion to be marked as "fully supported", there must be at least one method of achieving it without failure throughout the *entirety* of all processes and pages of the product; **and** in instances where that is not possible, an alternate component is provided that achieves the same purpose in an accessible way.

- The VPAT does not...

✗ List what parts of your product are not accessible.

✓ It lists which *accessibility criteria* are not supported (or are).

Thus, the VPAT is not an ideal method for tracking specific accessibility needs for a product when trying to bring it up to standard. (Spreadsheets and guided documents are better)

Level Data

Appendix

Appendix A: Accessibility Toolbox

Appendix B: Tool Tutorials

Appendix C: Helpful Accessibility Resources

Appendix A: Accessibility Toolbox

Level Data

Automated Scanners

- Axe DevTools
- WAVE extension
- Accessibility Insights
- Siteimprove Accessibility Checker

Code Checkers for Developers

- Axe DevTools Linter
- jsx-a11y-lint
- jest-aXe

Screen Readers

- NVDA screen reader
- Your computer's built-in screen reader

Element Inspection Tools

- ANDI bookmarklet (multi-functional)
- WebAIM Contrast Checker (contrast only)

- Choose at least one from each category for your work (as applicable to your position), selecting the ones that meet your needs and comfortability

More details about these tools can be found in the following Shared Services Confluence document:

Scanning Tools - Accessibility [PSST-103]

Appendix C: Helpful Accessibility Resources

LevelData

Quick References:

Fundamental concepts

- Read me first! Understand the importance of ARIA
- Building accessible components
- Implementing practices
- A list of accessible name information based on ARIA role
- A list of definitions of roles

Understanding WCAG 2.1

- A list of all WCAG 2.1 Criteria
- A list of only WCAG 2.1 AA Criteria
- All WCAG 2.1 Understanding Docs

Understanding WCAG 2.2

- A list of all WCAG 2.2 Criteria
- A list of only WCAG 2.2 AA Criteria
- All WCAG 2.2 Understanding Docs

Understanding W3 resources

- Understanding techniques provided by W3C

Check This Out!

W3C's Digital Accessibility Foundations Free Online Course

Level Data

Thank You